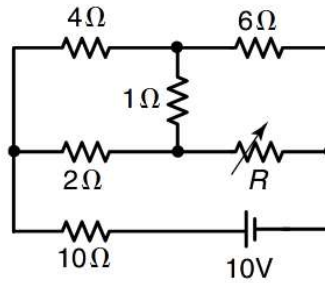
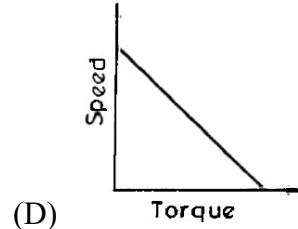
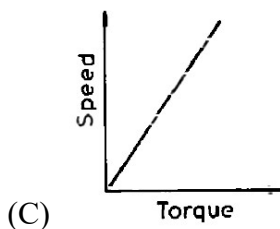
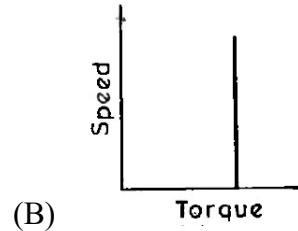
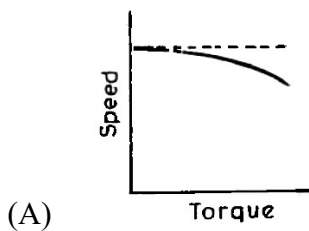


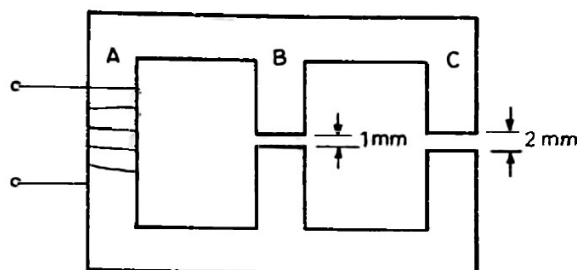
1. For the circuit shown in Fig. below, the current flowing through the 1 ohm resistor is adjusted to zero by varying the value of R. What is the value of R?



- (A) 2 ohm (B) 4 ohm (C) 6 ohm (D) 3 ohm
2. A voltage V is applied to an ac circuit resulting in the delivery of a current I . Which of the following expressions yield the true power delivered by the source?
1. Real part of $\bar{V}\bar{I}^*$
 2. Real part of VI
 3. I^2 times the real part of $\frac{\bar{V}}{\bar{I}}$
- Select the correct answer using the codes given below:
- (A) 1 and 3 (B) 1 only (C) 3 alone (D) 2 and 4
3. Under conditions of maximum power transfer from an ac source to a variable load,
- (A) The load impedance must also be inductive, if the generator impedance is inductive.
 - (B) The sum of the source and load impedance is zero.
 - (C) The sum of the source reactance and load reactance is zero.
 - (D) The load impedance has the same phase angle as the generator impedance.
4. The ratio of the readings of two wattmeters connected to measure power in a balanced 3-phase load is 5:3 and the load is inductive. The power factor of the load is
- (A) 0.92 lead (B) 0.92 lag (C) 0.60 lag (D) 0.60 lead
5. If the armature of a dc motor is supplied from a constant current source, and its field from a constant voltage source, then the speed-torque characteristics of the motor will be



6. The speed of a 3-phase IM is controlled by controlling its supply frequency. If the speed of the machine is reduced by reducing the frequency by 50% of the rated frequency; to keep the flux in the machine constant, the motor voltage compared to rated voltage must be
 (A) Increased by 25% (B) Increased by 50%
 (C) Decreased by 50% (D) Decreased by 25%
7. In the magnetic circuit shown in Fig. below, the areas of cross-section of limbs B and C are, respectively, 0.01m^2 and 0.02 m^2 . Air gaps of lengths 1.0 mm and 2.00 mm, respectively, are cut in limbs B and C. If the magnetic medium can be assumed to have infinite permeability and the flux in limb B is 1.0 Wb, the flux in the limb A is



- (A) 3 Wb (B) 1.5 Wb (C) 2 Wb (D) 4 Wb
8. Transformer maximum efficiency, for a constant load current, occurs at
 (A) At any power factor (B) Zero power factor leading
 (C) Zero power factor lagging (D) Unity power factor
9. A 4 kVA, 400/200 V, 50 Hz single phase transformer has leakage impedance of $0.02 + j 0.04$ per unit. This leakage impedance in ohms, when referred to h.v. side is
 (A) $0.8 + j 1.6 \Omega$ (B) $0.2 + j 0.4 \Omega$ (C) $0.08 + j 0.16 \Omega$ (D) $1 + j 2 \Omega$
10. A single-phase self-starting motor has two stator windings
 (A) Placed at 180° elect and fed with out-of-phase voltages
 (B) Placed at 90° elect and fed with in-phase voltages
 (C) Placed at 45° elect and fed with out-of-phase voltages
 (D) Placed at 90° elect and fed with out-of-phase voltages
11. Holes in an n-type semiconductor are
 (A) minority carriers that are thermally produced
 (B) minority carriers that are produced by doping
 (C) majority carriers that are thermally produced
 (D) majority carriers that are produced by doping
12. The dynamic resistance can be important; when a diode is
 (A) reverse-biased (B) forward-biased (C) in reverse breakdown (D) unbiased
13. A four-bit adder-subtractor can be constructed using four full adders and
 (A) Four two-input OR gates (B) Two four-input OR gates
 (C) Two four-input Exclusive-OR gates (D) Four two-input Exclusive-OR gates

14. To saturate a BJT,
 (A) $I_B = I_{C(sat.)}$ (B) $I_B > \frac{I_{C(sat.)}}{\beta_{DC}}$
 (C) V_{CC} must be at least 10 V (D) The emitter terminal must be grounded
15. The constant-current region of a FET lies between
 (A) cutoff and saturation (B) cutoff and pinch-off
 (C) 0 and I_{DSS} (D) pinch-off and breakdown
16. When an op-amp is operated in the single-ended differential mode,
 (A) the output is grounded
 (B) one input is grounded and a signal is applied to the other
 (C) both inputs are connected together
 (D) the output is not inverted
17. A phase-shift oscillator has
 (A) three RC circuits (B) three LC circuits (C) a T-type circuit (D) a π type circuit
18. A 48 kHz clock signal having a duty cycle of 25% is used to clock a four-bit binary ripple counter. What will be the frequency and duty cycle of true output of the MSB flip-flop?
 (A) 3 kHz, 25 % (B) 12 kHz, 25 % (C) 3 kHz, 50 % (D) 12 kHz, 50 %
19. In amplitude modulation
 (A) Bandwidth is infinite (B) Sideband power is always constant
 (C) Total transmitted power is constant (D) Carrier power is constant
20. The linear variable differential transformer transducer is
 (A) Resistive transducer (B) Inductive transducer
 (C) Capacitive transducer (D) Inductive-Capacitive Transducer
21. The output for the following C program will be:
- ```
void main() {
 int i, k, x = 0;
 for (i = 0; i < 5; ++i)
 for (k = 0; k < i; ++k) { x+= (i + k - 1); }
 printf("%d", x); }
```
- (A) 10 (B) 15 (C) 25 (D) 30
22. The output for the following C program will be:
- ```
void main() {
  int a = 5, b = 10;
  a = a + b; b = a - b; a = a - b;
  printf("%d , %d", a, b); }
```
- (A) 5, 10 (B) 15, 5 (C) 10, 5 (D) 0, 0

23. Encapsulation in Object Oriented Programming is a:
- (A) Concept that allows overriding of functions
 - (B) Concept that allows data hiding
 - (C) Concept that allows modular programming
 - (D) Concept that allows wrapping of data and functions in a single unit
24. The output for the following C program will be:
- ```
void main() {
 int a[5]={5, 1, 15, 20, 25};
 int i, j, m;
 i = ++a[1];
 j = a[1]++;
 m = a[i++];
 printf("%d%d%d", i, j, m); }
```
- (A) 1532                      (B) 3215                      (C) 2135                      (D) 1325
25. The output for the following C program will be:
- ```
void main() {
    int a[3] = {10,12,14}; a[1]=20; int i=0;
    while(i<3) { printf("%d ", a[i]); i++; } }
```
- (A) 20 12 14 (B) 10 20 14 (C) 10 12 20 (D) 12 20 14
26. The output for the following C program will be:
- ```
struct st
{
 int color;
}p1[2];
void main() {
 struct st p2[3];
 p1[0].color=5 + p2[0].color+ p1[1].color;
 p1[1].color=9 + p2[0].color + p1[0].color;
 printf("%d, ",p1[0].color);
 printf("%d",p1[1].color);
}
```
- (A) 5, 14                      (B) 5, 9                      (C) 19, 23                      (D) 23, 19
27. A mode which is used to open an existing file for both reading and writing \_\_\_\_\_
- (A) "W"                      (B) "W+"                      (C) "R+"                      (D) "A+"
28. Why is the & (address-of) operator used with scanf() when reading integer values?
- (A) To specify the format of the input.
  - (B) To store the input at the variable's memory address.
  - (C) To perform pointer arithmetic.
  - (D) It is optional and does not affect the function's behavior.

29. What will be the value of z if x = 8 after the execution of following statement:

```
void main() {
 int x=8,z,y; y = (x > 6 ? 4 : 6);
 z=(y > x ? 6 : 4);
 printf("%d %d",y,z);
}
```

- (A) 4 6                      (B) 4 4                      (C) 6 4                      (D) 6 6
30. How many times the program will print "PU"?
- ```
int main() {
    printf("PU");
    main();
    return 0;
}
```
- (A) Infinite times (B) 32767 times (C) 65535 times (D) Till stack overflows
31. The ratio of two specific heats of air is equal to
- (A) 0.17 (B) 0.24 (C) 0.1 (D) 1.41
32. The constant volume cycle is also called
- (A) Carnot cycle (B) Joule cycle (C) Diesel cycle (D) Otto cycle
33. Water boils when its vapour pressure
- (A) Equals that of the surroundings
 (B) Equals 760 mm of mercury
 (C) Equals to atmospheric pressure
 (D) Boiling has nothing to do with vapour pressure
34. While steam expands in turbines, theoretically the entropy
- (A) Remains constant (B) Increases
 (C) Decreases (D) Behaves unpredictably
35. The property of fluid by virtue of which it offers resistance to shear is called
- (A) Surface tension (B) Adhesion (C) Cohesion (D) Viscosity
36. The resultant upward pressure of the fluid on an immersed body due to its tendency to uplift the submerged body is called
- (A) Upthrust (B) Reaction (C) Buoyancy (D) Metacentre
37. Barometer is used to measure
- (A) Pressure in pipes (B) Atmospheric pressure
 (C) Rain level (D) Difference of pressure between two points
38. Hooke's law holds good upto
- (A) Yield point (B) Limit of proportionality
 (C) Breaking point (D) Elastic point

39. Stress in a beam and moment
 (A) Are directly proportional (B) Are inversely proportional
 (C) Have curvilinear relation (D) Have unpredictable relationship
40. A hollow shaft of same cross-section area and material as solid shaft transmits
 (A) Same torque (B) Less torque (C) More torque (D) Cannot be predicted
41. In the limit state approach, spacing of main reinforcement controls primarily
 (A) Cracking (B) Deflection (C) Durability (D) Collapse
42. If the fore bearing of a line AB is 35° and that of line BC 15° , then the included angle between the lines is
 (A) 20° (B) 50° (C) 160° (D) 230°
43. Number of vertical joints in a stretcher course is x times the number of joints in the header course, where x is equal to
 (A) $\frac{1}{4}$ (B) 1 (C) 2 (D) $\frac{1}{2}$
44. A sudden jump anywhere on the bending moment diagram of a beam is caused by:
 (A) Couple acting at that point (B) Couple acting at some other point
 (C) Concentrated load at the point (D) Uniformly varying load on the beam
45. Highway facilities are designed for
 (A) Annual average hourly volume (B) Annual average daily traffic volume
 (C) Thirtieth highest hourly volume (D) Thirteenth highest hourly volume
46. To make one cubic meter of 1:2:4 by volume concrete, the volume of coarse aggregates required is
 (A) 0.68 m^3 (B) 0.78 m^3 (C) 0.88 m^3 (D) 0.98 m^3
47. For a steel built-up column subjected to an axial force of 1200 kN, the lacing system is to be designed for resisting transverse shear of
 (A) 15 kN (B) 20 kN (C) 25 kN (D) 30 kN
48. The soils most susceptible to liquefaction are
 (A) Saturated dense sands
 (B) Saturated fine and medium sands of uniform particle size
 (C) Saturated clays of uniform size
 (D) Saturated gravels and cobbles
49. The fineness of cement is tested by
 (A) Air-content test (B) Air-permeability Method
 (C) Le-Chatelier apparatus (D) Vicat's apparatus
50. If A be the area of cross-section of a bar, the gauge length for the measurement of ductility will be
 (A) $5.65 \times A^{1/2}$ (B) $5.65 \times A^2$ (C) $6.56 \times A^{1/2}$ (D) $6.56 \times A$

51. A vacuum of 100 mm Hg is equivalent to an absolute pressure of
 (A) 100 mm Hg. (B) 660 mm Hg. (C) 860 mm Hg. (D) 100/760 mm Hg.
52. 1 kmol of glycerin contains
 (A) 1 k atom of carbon (B) 2 k atom of carbon
 (C) 3 k atom of carbon (D) 4 k atom of carbon
53. Which of the following increases with increase in specific gravity at 288.7/ 288.7 K
 (A) ° Twadell (B) ° Baume (C) ° API (D) ° Brix
54. Oxygen content of the flue gases on wet basis is determined by
 (A) Orsat Analyzer (B) Fyrite apparatus
 (C) Zirconium oxide cell (D) Calorimeter
55. A mixture of hydrogen and oxygen at ordinary atmospheric conditions cannot exist in
 (A) Thermal equilibrium
 (B) Mechanical equilibrium
 (C) Chemical equilibrium
 (D) Thermal, chemical and mechanical equilibrium
56. For any cyclic process
 (A) $\Delta U = 0$ (B) $Q = 0$
 (C) $W = 0$ (D) ΔU , Q and W are all zero
57. In a cyclic process, the work done by an ideal monoatomic gas is 400 J. The heat absorbed by the gas is
 (A) 0 (B) - 400 J (C) + 400 J (D) + 500 J
58. The point at which both liquid and gas phases are identical is called
 (A) Critical point (B) Triple point (C) Freezing point (D) Boiling point
59. The value of the ratio of the specific heat for monoatomic gas is
 (A) 1.66 (B) 1.44 (C) 1.55 (D) 1.89
60. The volume occupied by 1 kmol of an ideal gas at 273.15 K and 101.325 KPa is
 (A) 22.414 m³ (B) 359 m³ (C) 22414 m³ (D) 35.9 m³
61. In a simple Atwood's machine, two masses are connected by a light, inextensible string over a frictionless pulley. If $m_1=5$ kg and $m_2=3$ kg, calculate the acceleration of the system. Take the acceleration due to gravity $g=10$ m/s².
 (A) 3.5 m/s² (B) 2.5 m/s² (C) 1.5 m/s² (D) 0.5 m/s²
62. The velocity and acceleration of a particle performing simple harmonic motion have a steady phase relationship. The acceleration shows a phase lead over velocity of:
 (A) π (B) $\pi/2$ (C) $\pi/4$ (D) $-\pi$
63. The diameter of the objective lens of a telescope is 250 cm. For light with a wavelength of 540 nm coming from a distant object, the telescope's limit of resolution is
 (A) 2.64×10^{-7} radians (B) 1.54×10^{-7} radians
 (C) 3.5×10^{-7} radians (D) 4.64×10^{-7} radians

64. Two polaroids are crossed to each other. One of them is rotated through 45° , what % age of incident unpolarised light will pass through the system?
 (A) 50 % (B) 60% (C) 25% (D) 0%
65. Calculate the change in temperature of 1.0 kg of water that is initially at room temperature if 3.0 kJ of heat is added.
 (A) 358°C (B) 716°C (C) 0.36°C (D) 0.72°C
66. Heat transfer through a vacuum can occur by:
 (A) Conduction only (B) Convection only
 (C) Radiation only (D) Both conduction and convection
67. A capacitor has a capacitance C . When a dielectric of dielectric constant K is inserted, the new capacitance becomes:
 (A) C/K (B) $C+K$ (C) CK (D) $C-K$
68. Which of the following is *not* an application of eddy currents?
 (A) Ammeter (B) Dead beat galvanometer
 (C) Metal detectors (D) Induction furnace
69. If an electron and proton have the same de-Broglie wavelength then the relation between kinetic energy of the electron of proton is:
 (A) $KE_{\text{electron}} = KE_{\text{Proton}}$ (B) $KE_{\text{proton}} > KE_{\text{electron}}$
 (C) $KE_{\text{electron}} > KE_{\text{proton}}$ (D) $KE_{\text{electron}} = 2 KE_{\text{proton}}$
70. Zinc crystallizes in a hexagonal close-packed (hcp) structure. If r is the radius of a zinc atom, the height of the unit cell is
 (A) $2r\sqrt{\frac{8}{3}}$ (B) $r\sqrt{\frac{8}{3}}$ (C) $2r\sqrt{\frac{3}{8}}$ (D) $r\sqrt{\frac{3}{8}}$
71. Which of the following metals is not a pollutant?
 (A) Mercury (B) Arsenic (C) Lead (D) Aluminium
72. Among following, which one causes both depletion of ozone layer and green house effect?
 (A) SO_2 (B) CO_2
 (C) Chlorofluorocarbons (D) Water vapors
73. The end point of the complexometric determination of hardness of water using EDTA is
 (A) wine red to blue (B) wine red to colorless
 (C) blue to wine red (D) blue to colorless
74. The major impurity responsible for the scale formation in boilers is
 (A) CaCO_3 (B) CaSO_4 (C) MgCO_3 (D) Na_2SO_4
75. Which corrosion control technique is most suitable in the case of buried iron pipelines?
 (A) Anodic metal coating (B) Anodic protection
 (C) Cathodic protection (D) Corrosion inhibitors

76. The metal which can protect itself by forming a layer of its own oxide is
 (A) platinum (B) gold (C) iron (D) aluminium
77. What will be the net calorific value of a fuel which has 8% hydrogen? Given that the latent heat of steam and gross calorific value are 580 cal/g and 6500 cal/g respectively.
 (A) 7022 cal/g (B) 6022 cal/g (C) 7082 cal/g (D) 6082 cal/g
78. The porosity coke increases with increase in which of the following constituent of the coal from which it is prepared
 (A) Volatile matter content (B) Moisture content
 (C) Trace metals (D) Sulphur content
79. Modern periodic table is based on the
 (A) atomic mass (B) mass number (C) atomic number (D) atomic volume
80. The correct sequence of electron affinity of C, N, O and F is
 (A) $C > N < O > F$ (B) $C > N < O < F$ (C) $C < N > O < F$ (D) $C > N > O > F$
81. If one root of the equation $3x^2 + kx + 81 = 0$ (having real roots) is the square of the other root then find the value of 'k'.
 (A) 3 (B) 2 (C) 4 (D) -1
82. Consider the following system of equations $AX = B$, where A is an $m \times n$ matrix, B is an $m \times 1$ matrix and X is the variable vector of dimension $n \times 1$. Assuming that the system is consistent, what is the condition for the existence of infinitely many solutions. Given that $m < n$ and $[A:B]$ denotes the augmented matrix of the system.
 (A) $\text{Rank}(A) = \text{Rank}([A:B]) = n$ (B) $\text{Rank}(A) = \text{Rank}([A:B]) < m$
 (C) $\text{Rank}(A) \neq \text{Rank}([A:B])$ (D) $\text{Rank}(A) = \text{Rank}([A:B]) > n$
83. Find the value of $\cos(15^\circ)$.
 (A) $\sqrt{3}/2\sqrt{2}$ (B) $(1 - \sqrt{3})/2\sqrt{2}$ (C) $(1 + \sqrt{3})/2\sqrt{2}$ (D) $1/2\sqrt{2}$
84. Find the angle between the straight lines having equations $3x+4y-10=0$ and $4x-5y+2=0$.
 (A) $\tan^{-1}\left(\frac{30}{7}\right)$ (B) $\tan^{-1}(27)$ (C) $\tan^{-1}\left(\frac{31}{8}\right)$ (D) $\pi/4$
85. The n-th derivative of $y(x) = e^x(2x + 3)^3$ is _____.
 (A) $e^x[(2x + 3)^3 + 6n(2x + 3)^2 + 12n(n - 1)(2x + 3) + 8n(n - 1)(n - 2)]$
 (B) $e^x[(2x + 3)^3 - 6n(2x + 3)^2 + 12n(n - 1)(2x + 3) - 8n(n - 1)(n - 2)]$
 (C) $e^x[(2x + 3)^3 + n(2x + 3)^2 + n(n - 1)(2x + 3) + n(n - 1)(n - 2)]$
 (D) $e^x[(2x + 3)^3 - n(2x + 3)^2 + n(n - 1)(2x + 3) - n(n - 1)(n - 2)]$
86. Find the linearization of the function $f(x, y, z) = x^2 - xy + 3 \sin z$ at the point P(2,1,0).
 (A) $L(x,y,z) = 3x+2y+z$ (B) $L(x,y,z) = x-2y+3z-3$
 (C) $L(x,y,z) = 3x-2y+3z-2$ (D) $L(x,y,z) = 2x-3y+z-4$
87. Find the general solution of the differential equation $2(y - 4x^2)dx + x dy = 0$.
 (A) $x^2y + 2x^4 = c$ (B) $x^2y - 2x^4 = c$
 (C) $xy - x^4 = c$ (D) $xy^2 - x^4 = c$

88. Find the area of the region enclosed by the positive x-axis and spiral $r = 4\frac{\theta}{3}$, $0 \leq \theta \leq 2\pi$. The region looks like a snail shell.
 (A) $2\pi^2$ (B) $\frac{64}{27}\pi^3$ (C) $\frac{8}{9}\pi^3$ (D) π
89. Find the torsion for the space curve $\vec{r}(t) = (t^3/3)\hat{i} + (t^2/3)\hat{j} + 3t\hat{k}$, $t > 0$ at $t=1$.
 (A) 1 (B) 2 (C) -1 (D) 0
90. Using Divergence Theorem find the outward flux of the field $\vec{F} = (y-x)\hat{i} + (z-y)\hat{j} + (y-x)\hat{k}$ across the boundary of the region D: The cube bounded by the planes $x = \pm 1, y = \pm 1, z = \pm 1$.
 (A) -16 (B) 16 (C) 10 (D) 0
91. In a certain code language, CHECK is written as 6165622 and SURGE is written as 382136145. How will PRIEST be written as in that language?
 (A) 1618953840 (B) 3236953840 (C) 3436953640 (D) 3236185384
92. The distance between A and B is 20m and that between C and D is 40m. C is to the west of D, which is to the east of A at a distance of 50m. B is to the north of C. In which direction is D with respect to B?
 (A) North-east (B) South-west (C) South-east (D) North-west
93. Given below is a statement followed by two conclusions. Read them carefully and decide which of the conclusion(s) logically follows the given statement(s).
Statements:
 The quality of teachers in government schools has deteriorated and parents prefer to enroll their children in private schools.
Conclusion:
 I) The authorities should take immediate action to improve the quality of teachers in schools.
 II) All government schools should either be privatized or closed completely.
 (A) Only conclusion I follows
 (B) Only conclusion II follows
 (C) Either conclusion I or conclusion II follows
 (D) Neither conclusion I nor conclusion II follows
94. This is a letter
 P : from a young lady
 Q : who was lately wounded in a duel
 R : written in the most passionate terms
 S : wherein she laments the misfortune of a gentleman
 The Proper sequence should be:
 (A) PRSQ (B) PRQS (C) RPQS (D) RPSQ
95. Directions: The following sentences has word(s) underlined. Read the sentence carefully and find which word class the underlined word(s) belong(s) to. Indicate your response.
I don't really want to go. Besides, it's too late now
 (A) Adjective (B) Adverb (C) Preposition (D) Conjunction

96. Select the most appropriate option to fill in the blank.
The old Neem tree which _____ in our compound for fifty years suddenly crashed to the ground last night.
(A) had stood (B) has stood (C) is standing (D) stands
97. Select the option that expresses the given sentence in passive voice.
One should keep one's promises.
(A) Promises should be kept. (B) One should keep their promises.
(C) One's promises should be kept by one. (D) Promises should be kept by one.
98. Which of the following is an example of Indirect Personal Pronoun, reflexive?
(A) They
(B) to -myself
(C) myself
(D) More than one of the above
99. Select the pair which has the same relationship.
THRUST:SPEAR
(A) mangle:iron (B) scabbard:sword (C) bow:arrow (D) fence:epee
100. Direction: Choose the correct form of the verb that is in agreement with the subject:
The mother, along with her daughter, _____ tennis on Sundays.
(A) has played (B) have playing (C) play (D) plays

x-x-x

(ROUGH WORK)

(ROUGH WORK)

(ROUGH WORK)

(ROUGH WORK)